

Task - RealPlay Backend Engineer

We run a **NestJS/Fastify** backend with **Postgres (Prisma ORM)**, **Redis**, and a separate **workers** app.

Goal: a small **Tournaments** NestJS module

Build a small NestJS service that creates tournaments, ingests bet events, exposes a live leaderboard, and writes final results after the tournament ends.

Required endpoints

- **POST /bet** - processes bets from client
- **POST /tournaments** - create a tournament with **name**, **startsAt**, and **endsAt**
- **GET /tournaments/:id/leaderboard?limit&offset** - return paginated player placements on leaderboard sorted by score **DESC**

Bet Payload

```
{
  "externalBetId": "bet_123456", // string
  "playerId": "player_42", // string
  "amount": 250, // int
  "currency": "USD", // string
  "createdAt": "2026-06-04T12:30:00.000Z" // string (date)
}
```

Rules

- **amount** is in cents, so 250 means \$2.50.
- A bet counts only when **startsAt** <= **createdAt** <= **endsAt**.
- A bet can count towards more than **one** active tournament.
- Each **externalBetId** must be counted only once per tournament. Duplicate events should return success/idempotent response without increasing score twice.
- Tournament score is the running sum of accepted bet amounts per player.
- Use **Redis** for live ranking
- Use **BullMQ** to schedule snapshot job to write final placements to Postgres after **endsAt**

Deliverables

- A runnable service with clear setup instructions.
- Prisma schema and migration/s.
- A short README explaining assumptions and tradeoffs.
- (BONUS) Minimal tests covering bet ingestion, duplicate bet handling, and leaderboard ordering.